GEOGRAPHY

INDIAN GEOGRAPHY





INDIAN SOILS

MODULE - 7

- PROBLEM OF INDIAN SOIL
- SALINE & ALKALINE SOIL

MAJOR SOIL TYPES OF INDIA

SALINISATION & ALKALISATION

WHAT DO YOU MEAN BY SALINISATION?

1. It is the process of accumulation of salts, such as sulphates and chlorides of calcium, magnesium, sodium and potassium, in soils in the form of a salty horizon.
2. It is quite common in arid and semi arid regions. It may also take place through capillary rise of saline ground water and by inundation with seawater in marine and coastal soils. Salt accumulation may also result from irrigation or seepage in areas of impeded drainage.

WHAT DO YOU MEAN BY ALKALISATION?

1. The process involves the accumulation of sodium ions resulting in the formation of sodic soils is known as alkalization.
2. Alkalinity implies the dominance of sodium salts, specially sodium carbonate.

MAJOR SOIL TYPES OF INDIA

8 SALINE & ALKALINE SOILS

Salinity creeping into croplands

1. In Saline and Alkaline Soils, the top soil is impregnated (soak or saturate with a substance) with saline and alkaline efflorescence (become covered with salt particles).
2. There are many undecomposed rock and mineral fragments which on weathering liberate sodium, magnesium and calcium salts and sulphurous acid.
3. Some of the salts are transported in solution by the rivers which percolate in the sub-soils of the plains.

MAJOR SOIL TYPES OF INDIA

8 SALINE & ALKALINE SOILS

CAPILLARY ACTION

- *Capillary action is the ability of a liquid to flow in narrow spaces without the assistance of, and in opposition to, external forces like gravity.*
- *The force behind capillary action is surface tension.*

4 In regions with low water table, the salts percolate into sub soil and in regions with good drainage, the salts are washed away by flowing water.

5 But in places where the drainage system is poor, the water with high salt concentration becomes stagnant and deposits all the salts in the top soil once the water evaporates.

6 In regions with high sub-soil water table, injurious salts are transferred from below by the capillary action as a result of evaporation in dry season.

MAJOR SOIL TYPES OF INDIA

8 SALINE & ALKALINE SOILS

DISTRIBUTION OF SALINE SOILS

1. Saline and Alkaline Soils occupy 68,000 sq km of area.

2. These soils are found in canal irrigated areas and in areas of high sub-soil water table.

3. Parts of Andhra Pradesh, Telangana, Karnataka, Bihar, Uttar Pradesh, Haryana, Punjab (side effects of improper or excess irrigation), Rajasthan and Maharashtra have this kind of soils.

4. The accumulation of these salts makes the soil infertile and renders it unfit for agriculture.

5. In Gujarat, the areas around the Gulf of Khambhat are affected by the sea tides carrying salt-laden deposits. Vast areas comprising the estuaries of the Narmada, the Tapi, the Mahi and the Sabarmati have thus become infertile.

6. Along the coastline, saline sea waters infiltrate into coastal regions during storm surges (when cyclones make landfall) and makes the soil unfit for cultivation.

7. The low lying regions of coastal Andhra Pradesh and Tamil Nadu face this kind of soil degradation.

PROBLEMS OF INDIAN SOILS

Salinity creeping into croplands

1. Indian soils suffer from a number of problems, some of them taking a serious turn.

2. Some of the important problems faced by Indian soils are :

   I. Soil erosion
   II. Deficiency in fertility
   III. Desertification
   IV. Water logging
   V. Salinity and alkalinity
   VI. Wasteland
   VII. Over exploitation of soils due to increase in population and rise in living standards
   VIII. Encroachment of agricultural land due to urban and transport development.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

1. Soil erosion is the removal of soil by the forces of nature, particularly wind and water, more rapidly than the various soil forming processes that can replace it.
2. Soil erosion is a serious menace which is adversely affecting our agricultural productivity and the economy of the country as a whole.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

1 WATER EROSION

1. During heavy rains, water removes a lot of soil. The splash of the raindrops will detach particles of sand and silt most readily.
2. Coarser particles are not shifted about as much because of their greater volume and weight.
3. Run-off water is responsible for much of the soil erosion, moving the soil particles by surface creep, saltation and suspension.
4. Water erosion process mainly consists of rilling, gullying, sheet-wash and rain peeling process.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

1 WATER EROSION

5. If erosion continues unchecked for a sufficient time, numerous finger-shaped grooves may develop all over the area due to silt laden run off.

6. The whole pattern resembles that of the twigs, branches and trunk of a tree. This is called rill erosion.

7. With further erosion of the soil, the rills may deepen and become enlarged and are ultimately turned into gullies.

8. When a gully bed cuts into the soil with an immediate drop of 3 to 4 metres and gradually flattens out, a ravine is formed.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

1 WATER EROSION

Rill Erosion

Gully Erosion

Ravine Erosion

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

1 WATER EROSION

Sheet Erosion

9 The depth of a ravine may extend to 30 metres or even more.

10 When the entire top sheet of the soil is carried away by water or by wind, leaving behind barren rock, it is called sheet erosion.

11 Sheet erosion is no less harmful than gully erosion. This type of erosion is more prominent on relatively steeper slopes, receiving heavy rainfall.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

1 WATER EROSION

12 A single heavy downpour continuing for a few hours may result in severe soil erosion, while the same amount of rainfall fairly distributed over a longer period may cause little erosion or even may be useful for protecting soil.

13 The slope of the land is a potent factor in determining the velocity of water and the consequent soil erosion.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

1 WATER EROSION

Coastal Erosion

14 Other things being equal, the steeper the slope the more rapidly does water run down resulting in more soil erosion.

15 In the coastal areas, tidal waves dash along the coast and cause heavy damage to soil. This is called sea erosion.

16 In the higher reaches of the Himalayan region, soil erosion on a large scale is caused by glaciers.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

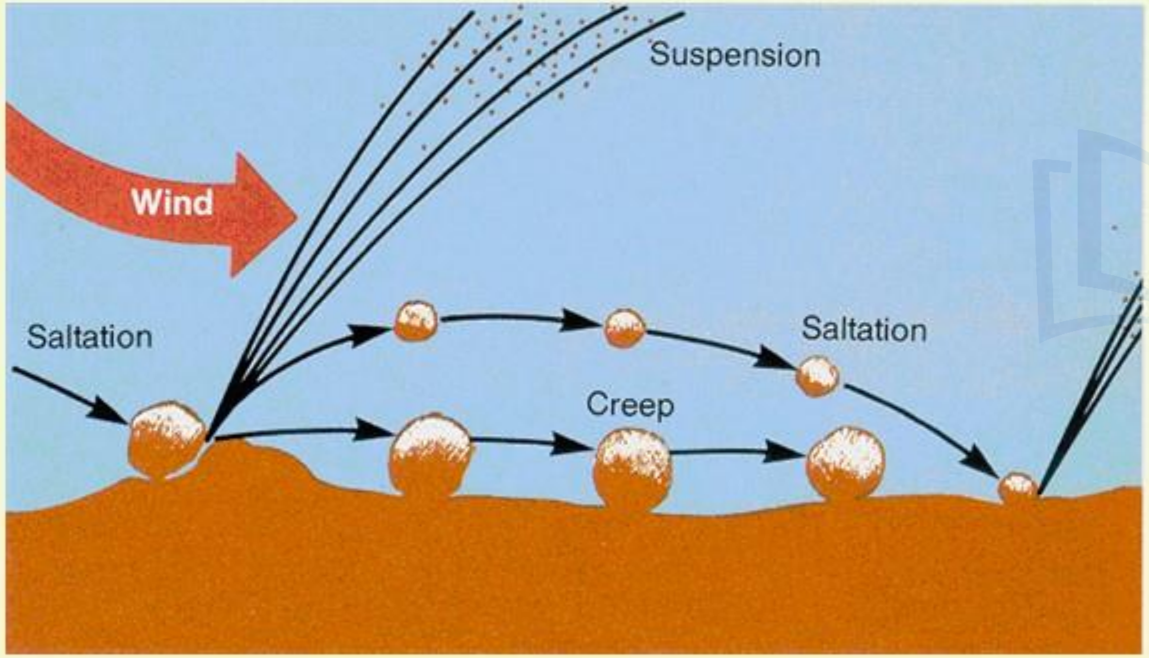
2 WIND EROSION

1. In arid and semi arid lands with little rainfall, the wind acts as a powerful agent of soil erosion causing heavy loss to agricultural land.
2. Winds blowing at considerable speed, remove the fertile, arable, loose soils leaving behind a depression devoid of top soil.
3. Wind erosion is accentuated when the soil is dry, weakly aggregated, devoid of vegetation cover along with over grazing and the winds are strong.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

2 WIND EROSION



Saltation

4. Even modest wind velocities can keep individual particles of humus, clay and silt in suspension.

5. Very fine, fine and medium sands are moved by wind in a succession of bounds and leaps, known as saltation

6. Coarse sand is not usually airborne but rather is rolled along the soil surface. This type of erosion is called surface creep.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

3 HUMAN FACTORS OF SOIL EROSION

Ploughing to be done across the slope

1. If the fields are ploughed along the slope, there is no obstruction to the flow of water and the water washes away the top soil easily.
2. In some parts of the country, the same crop is grown year after year which spoils the chemical balance of the soil.
3. This soil is exhausted and is easily eroded by wind or water.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

3 HUMAN FACTORS OF SOIL EROSION

4 Another outstanding example of faulty method of agriculture is the shifting cultivation practised in some areas in the north-eastern states of Arunachal Pradesh, Assam, Meghalaya, Manipur, Tripura, Mizoram, as well as in Orissa.

5 The removal of the forest cover leads to the exposure of the soil to rains and sun which results in heavy loss of top soil, especially on the hill slopes.

6 Thus the soil becomes unfit for cultivation and the tribes move to another piece of land after 2-3 years, returning to the earlier one after a gap of 10-15 years.

7 In this way, the whole of the forest area is adversely affected by shifting cultivation resulting in intensive soil erosion in vast areas.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

EXTENT OF SOIL EROSION IN INDIA

1. It has been estimated that about one-fourth of our total area is exposed to wind and water erosion out of which 40 million hectares of land has undergone serious erosion.

2. Wind erosion is a serious problem in arid and semi-arid parts of north west India.

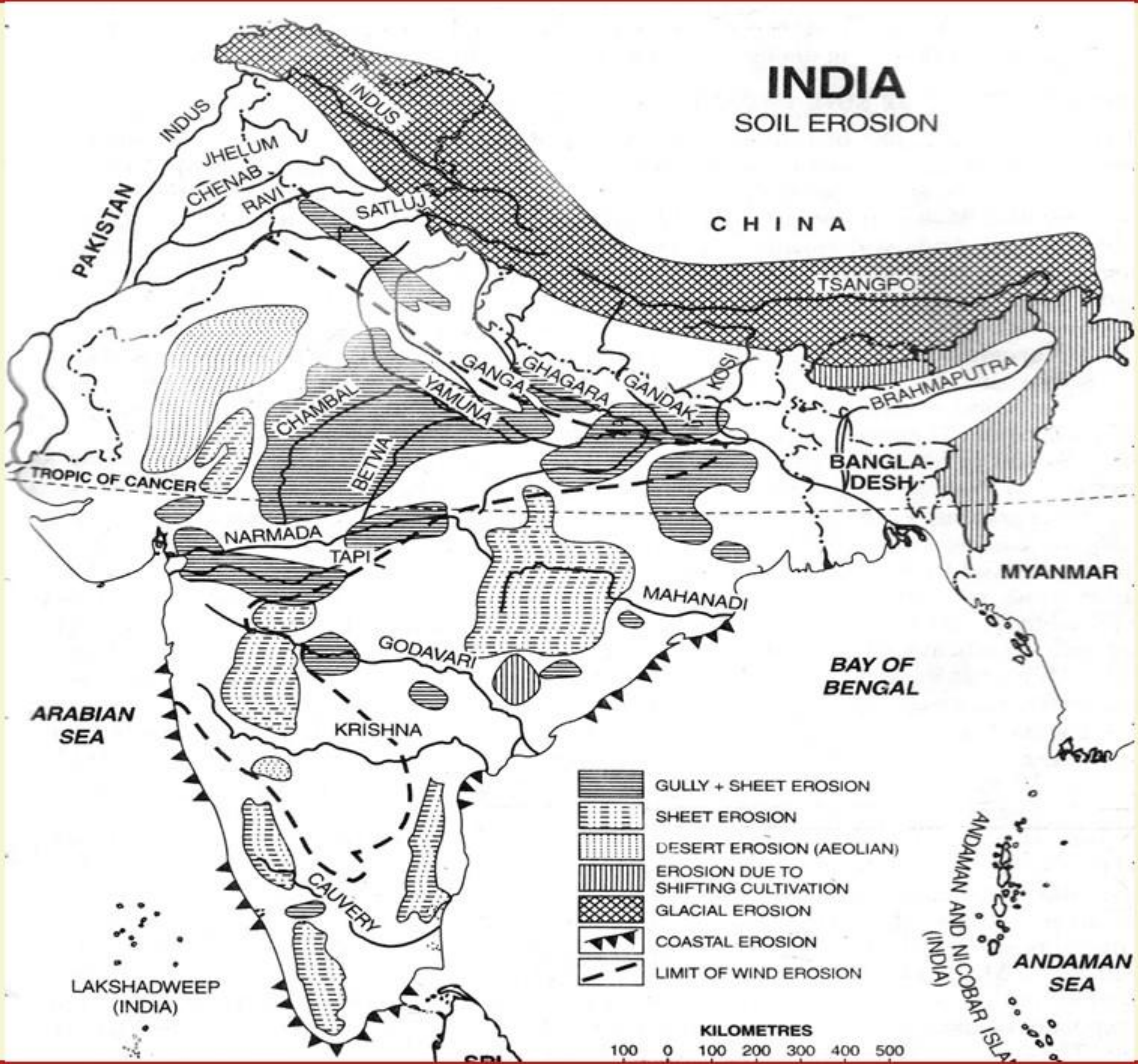
3. About 45 million hectares of land is subject to severe wind erosion in Rajasthan and adjoining areas of Punjab, Haryana, Gujarat and Western Uttar Pradesh.

4. They are devoid of vegetation cover and have sandy soil. As such these areas are exposed to wind erosion.

PROBLEMS OF INDIAN SOILS

SOIL EROSION

EXTENT OF SOIL EROSION IN INDIA



PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

EXTENT OF SOIL EROSION IN INDIA

Overgrazing is a major issue

5. Faulty farming practices, failure to conserve moisture, lack of managing and over grazing make the problem of wind erosion more complicated.
6. Water erosion is more active in wet areas receiving more rainfall.
7. Steep slope, swift flow of rivers and scarce vegetation cover lead to water erosion.
8. There are 39.75 lakh hectare ravines spread in 18 States, out of which 27.65 lakh hectare (or 69.55 per cent) are in the States of Uttar Pradesh, Madhya Pradesh, Rajasthan and Gujarat.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

EXTENT OF SOIL EROSION IN INDIA

Chambal ravines

9 In Madhya Pradesh about 4 to 8 lakh hectares are affected by deep gullies and ravines along the banks of rivers Chambal and Kali Sindh.

10 The flood plains of the Ganga and its tributaries in Uttar Pradesh and Bihar also suffer from the problems of soil erosion caused by water.

11 These rivers are carving deep furrows and removing fertile top soil.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

EXTENT OF SOIL EROSION IN INDIA

Chos of Punjab overflowing

12 The Shiwalik range has also been badly affected by gully erosion.

13 Rivers descending from the Shiwalik hills and flowing through Punjab are locally called Chos. Erosion by chos is most marked in Hoshiarpur district of Punjab.

14 In 130 km of the Shiwalik, nearly a hundred streams debouch onto the plains.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

EXTENT OF SOIL EROSION IN INDIA

Coastal Erosion

16 Coastal erosion is quite pronounced in the season of monsoon winds and during storms and cyclones.

17 Several coastal areas in Gujarat, Maharasthra, Karnataka, Kerala, Tamil Nadu, Andhra Pradesh and Orissa have suffered heavily at the hands of sea erosion.

PROBLEMS OF INDIAN SOILS

1 SOIL EROSION

EFFECTS OF SOIL EROSION

The adverse effects of soil erosion are reflected in the following points:

1. Top soil is eroded which leads to loss of soil fertility and fall in agricultural productivity.
2. Flooding and leaching result in loss of mineral nutrients.
3. Ground water level is lowered and there is decrease in soil moisture.
4. Natural vegetation covers dries up and arid lands expand.
5. Frequency and intensity of floods and drought increases.
6. Rivers, canals and tanks are silted and their water holding capacity decreases.
7. The incidence and damaging power of landslides increases.
8. Economy as whole suffers a great setback.

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OUR TEAM

G. V. Rao	Amrita Naidu
K. Srikanth	M. Binukrishna
D. Sunil Kumar	G. Ravi Babu
L. V. Krishna	D. Joji
Y. Deepthi	K. Victor Babu